

Impact of Node Density and Propagation Models on Wi-Fi Network Performance Using ns-3

Scalability, Reliability, and PHY-Layer Performance Analysis in Dense Wireless Networks.



NODE DENSITY



PROPAGATION MODELS



WI-FI NETWORK PERFORMANCE



SIMULATION WITH NS-3



SCALABILITY



RELIABILITY

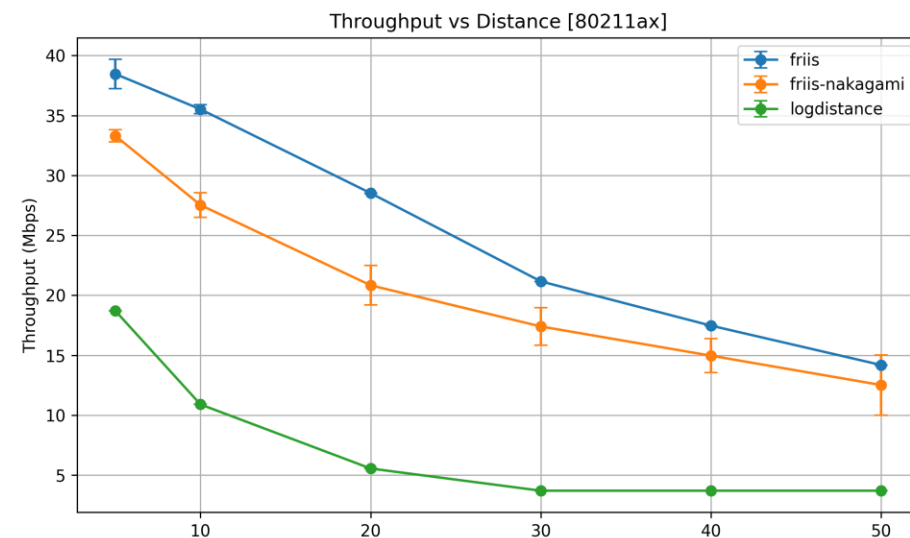
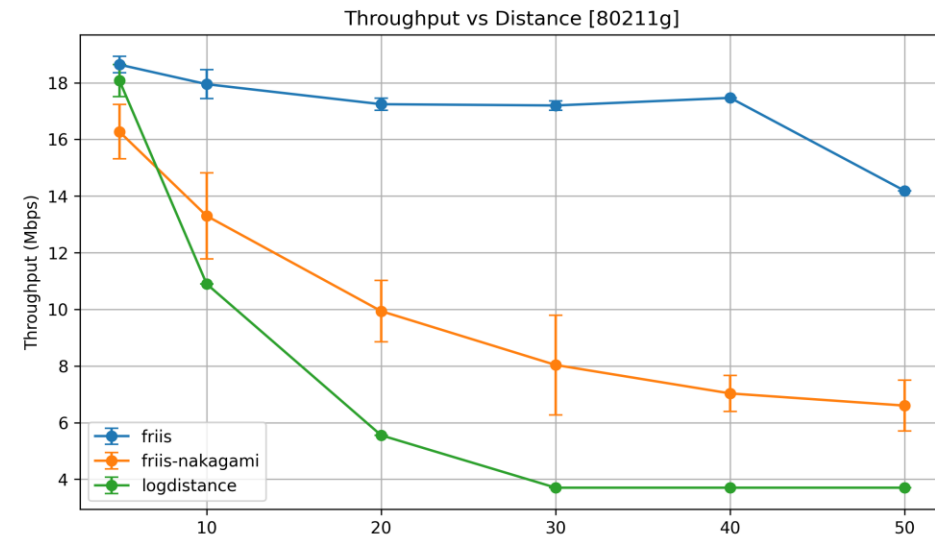


PHY-LAYER PERFORMANCE

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- **Presentation Structure**
- Research motivation
- Experimental methodology
- Throughput analysis
- Packet Delivery Ratio (PDR) analysis
- Delay and scalability analysis
- Conclusions



This work evaluates the impact of propagation models, node density, communication distance, and PHY-layer standards on wireless network performance using ns-3 simulations. A large-scale experimental campaign comprising 540 simulation runs was conducted to compare IEEE 802.11g and IEEE 802.11ax under Friis, LogDistance, and Friis-Nakagami propagation environments.

Experimental Methodology

Simulation Environment

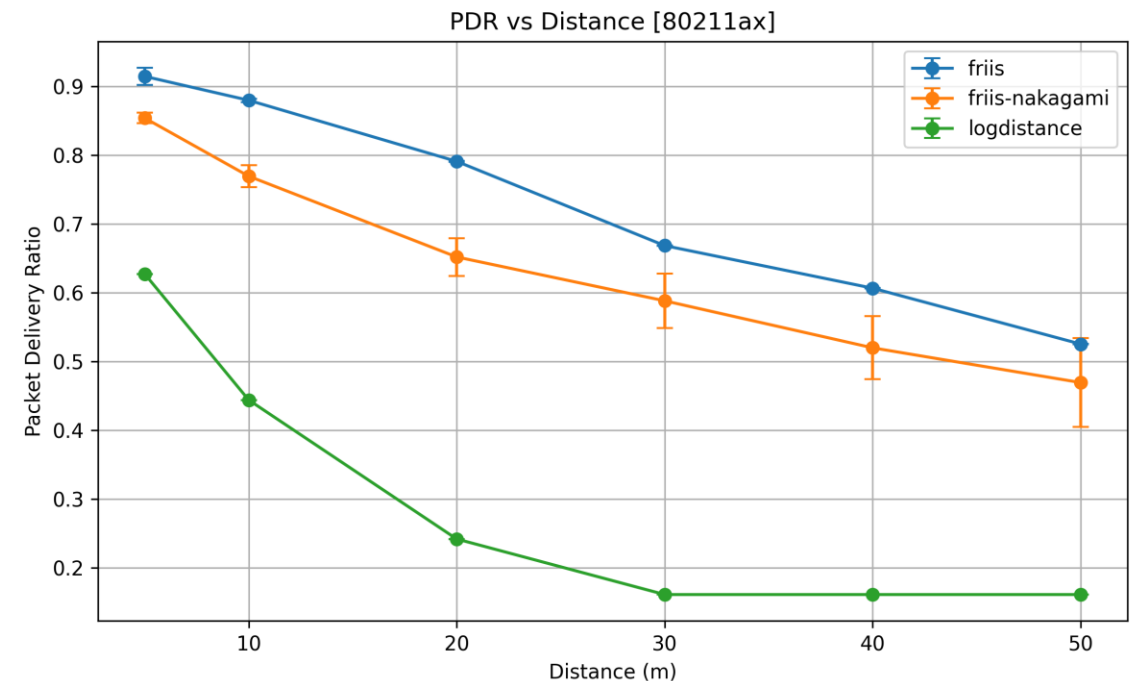
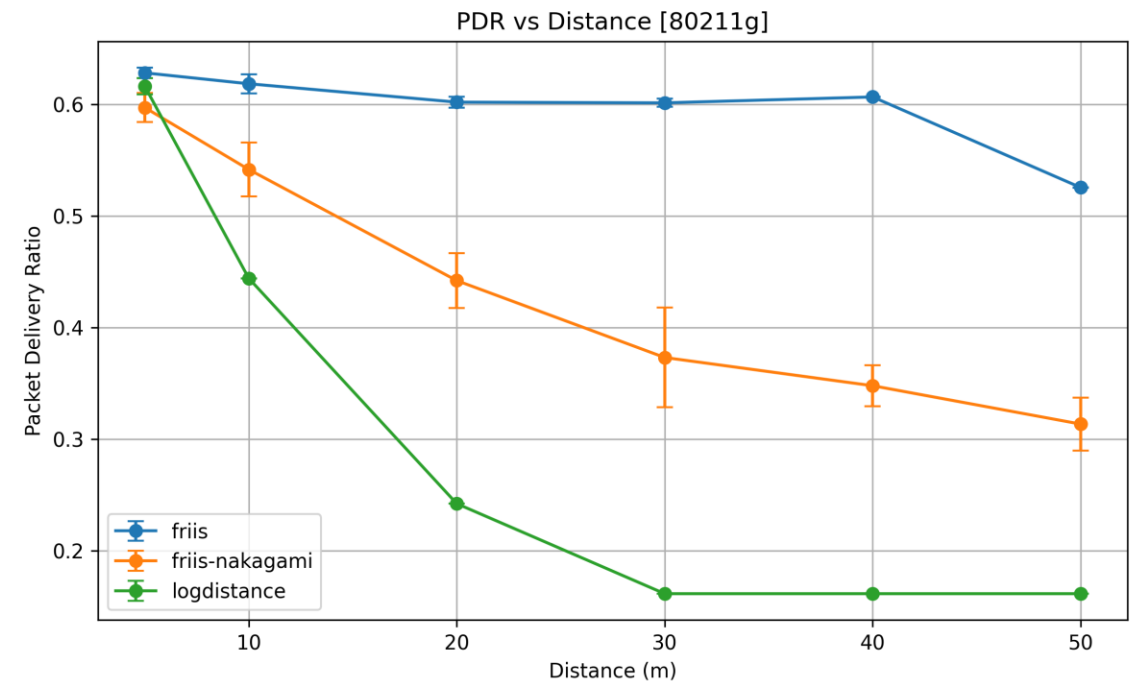
- Simulator: ns-3
- Traffic type: UDP
- Standards evaluated:
 - IEEE 802.11g
 - IEEE 802.11ax
- Propagation models:
 - Friis
 - LogDistance
 - Friis-Nakagami
- Node densities:
 - 5, 10, 20, 30, 50 nodes
- Communication distances:
 - 5 m to 50 m
- Multiple random seeds for statistical robustness

Evaluated Metrics

- Packet Delivery Ratio (PDR)
- Throughput
- End-to-End Delay
- Fairness
- Scalability

Scientific Observation

- The experimental matrix was designed to evaluate the combined effects of propagation degradation, network density, and PHY-layer capabilities on WLAN performance.



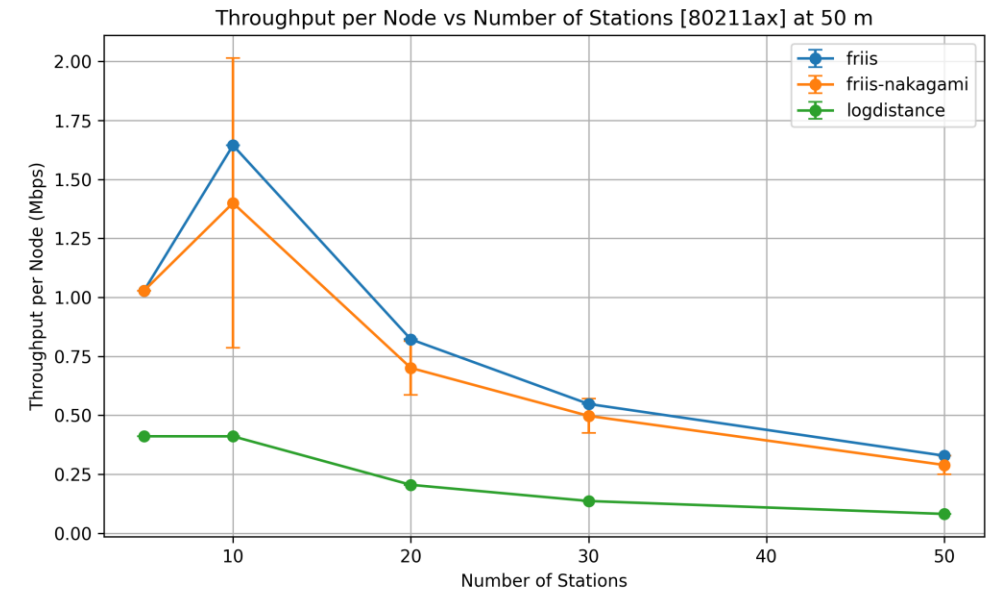
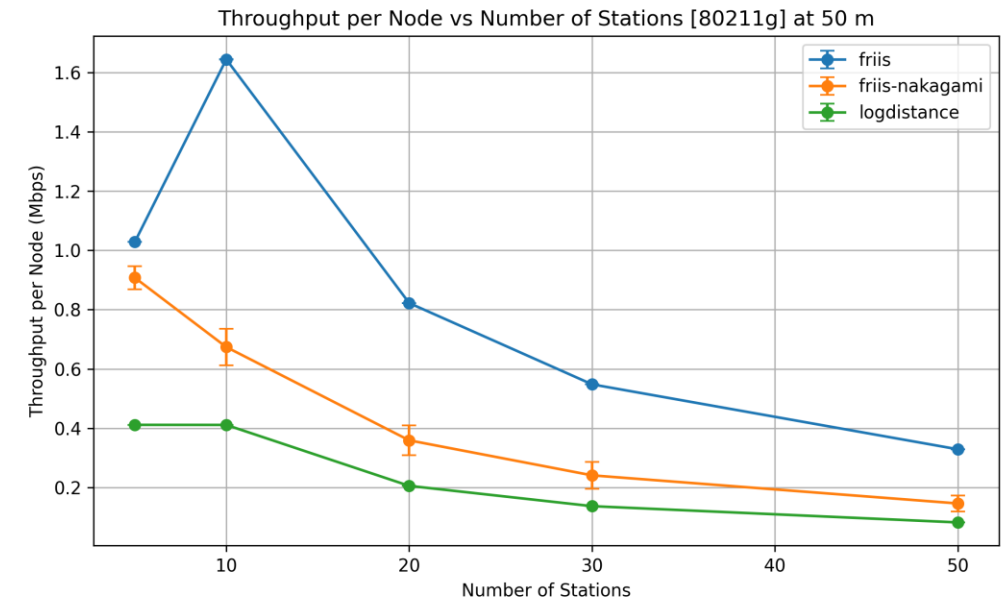
Throughput Analysis

Key Findings

- Throughput decreases as communication distance increases.
- Dense scenarios introduce channel contention and collision effects.
- IEEE 802.11ax consistently outperformed IEEE 802.11g.
- Friis propagation achieved the highest throughput values.
- LogDistance propagation produced severe throughput degradation.

Interpretation

- The results demonstrate that PHY-layer enhancements introduced by IEEE 802.11ax improve spectral efficiency and network capacity, particularly in dense deployment scenarios. However, scalability limitations remain visible under high-density conditions.



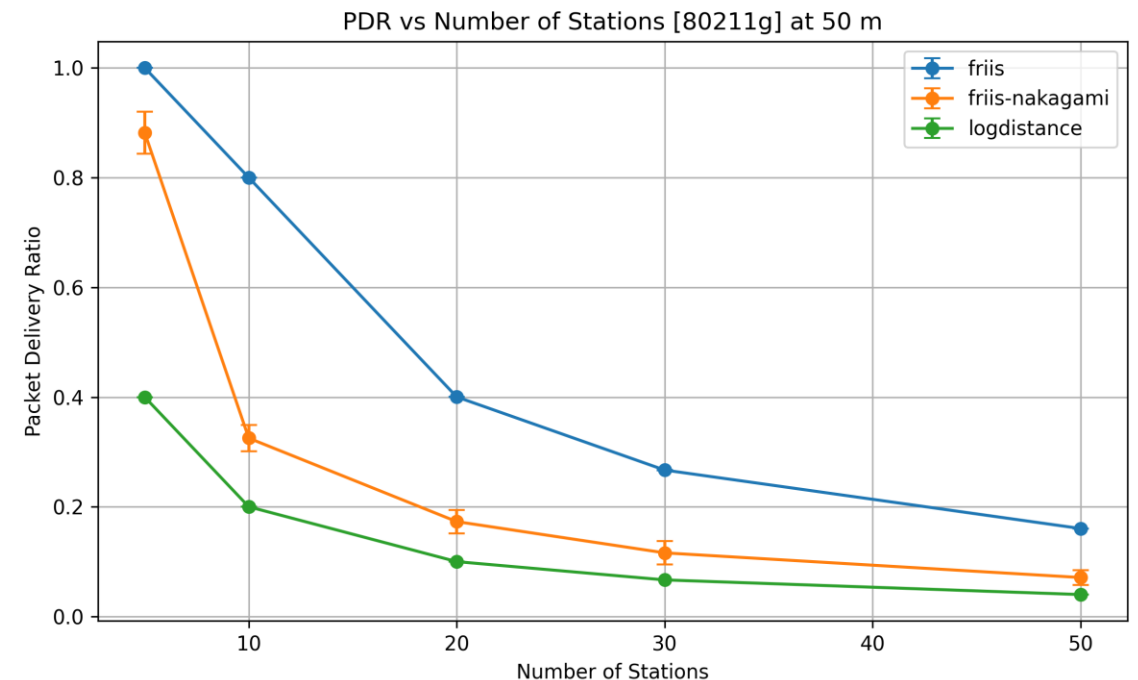
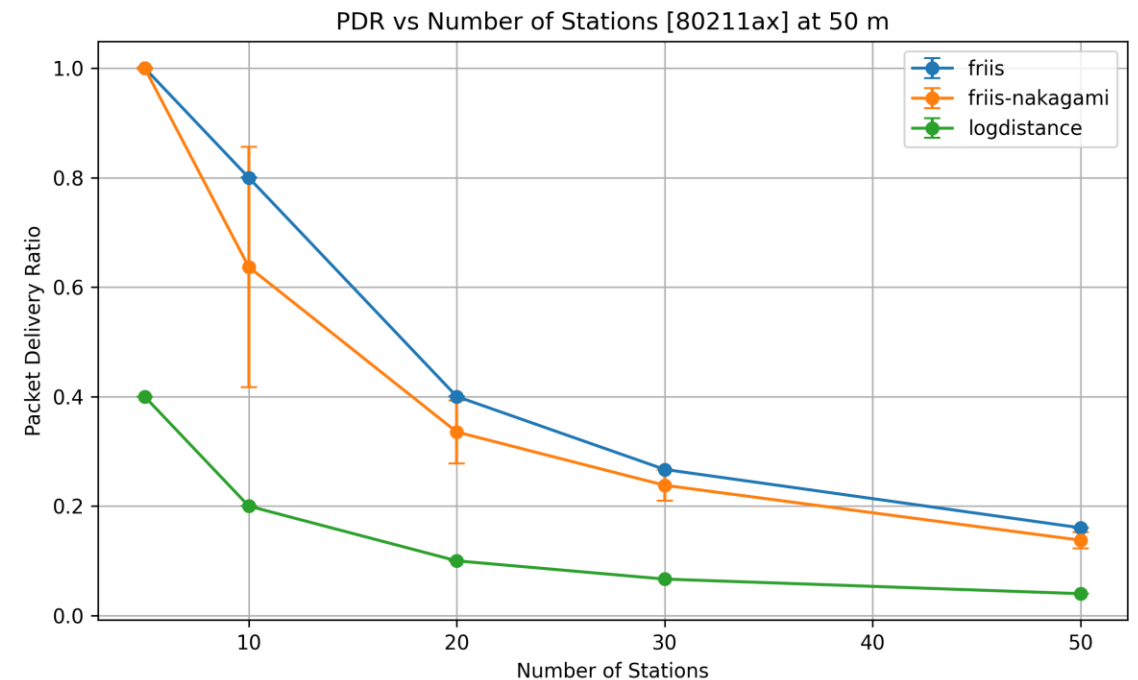
Packet Delivery Ratio (PDR) Analysis

Key Findings

- PDR decreases significantly with:
 - increasing distance
 - increasing node density
- Friis-Nakagami introduced strong fading variability.
- LogDistance showed the lowest reliability levels.
- IEEE 802.11ax maintained higher delivery reliability across all scenarios.

Scientific Interpretation

PDR results confirm that wireless scalability is strongly constrained by propagation attenuation and medium contention. IEEE 802.11ax improves robustness but cannot fully eliminate degradation in highly congested networks.



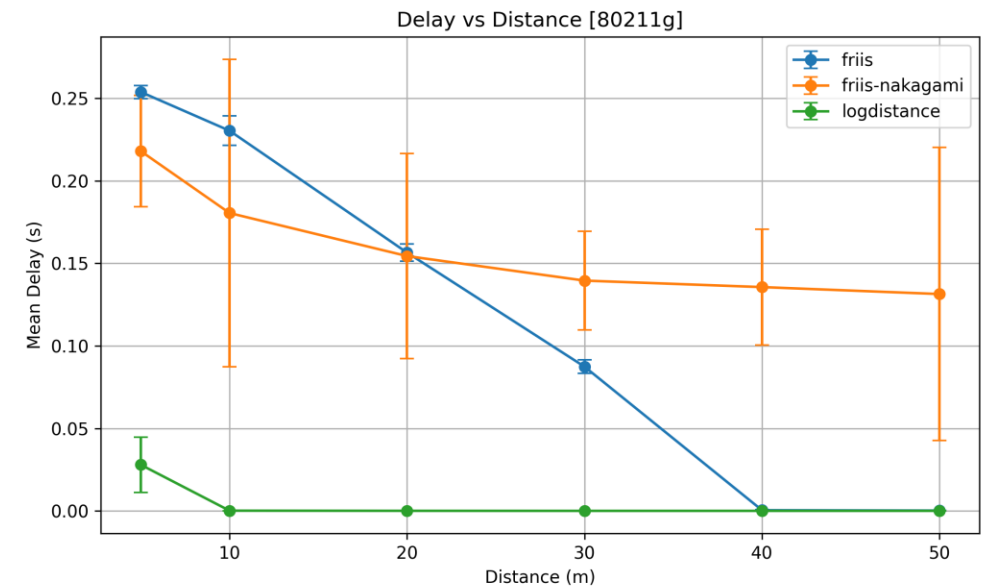
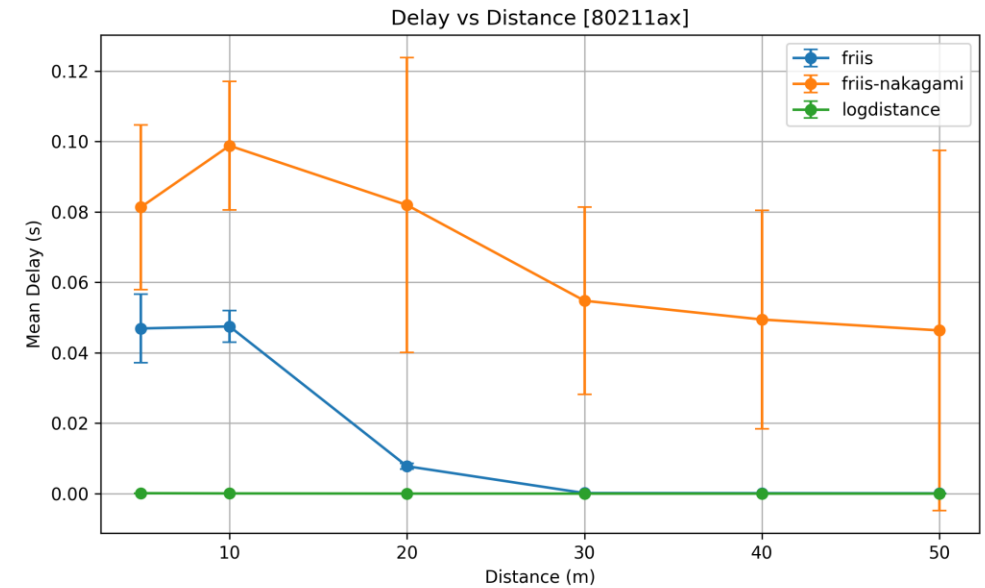
Delay and Scalability Analysis

• Key Findings

- Delay increases under dense traffic conditions.
 - Channel contention becomes dominant at high node densities.
-
- Friis-Nakagami generated the highest delay variability.
 - IEEE 802.11ax reduced average delay compared to IEEE 802.11g.

• Scientific Interpretation

- The delay analysis demonstrates the impact of MAC-layer contention and propagation instability on network responsiveness. Realistic fading environments substantially increase temporal instability in WLAN communications.



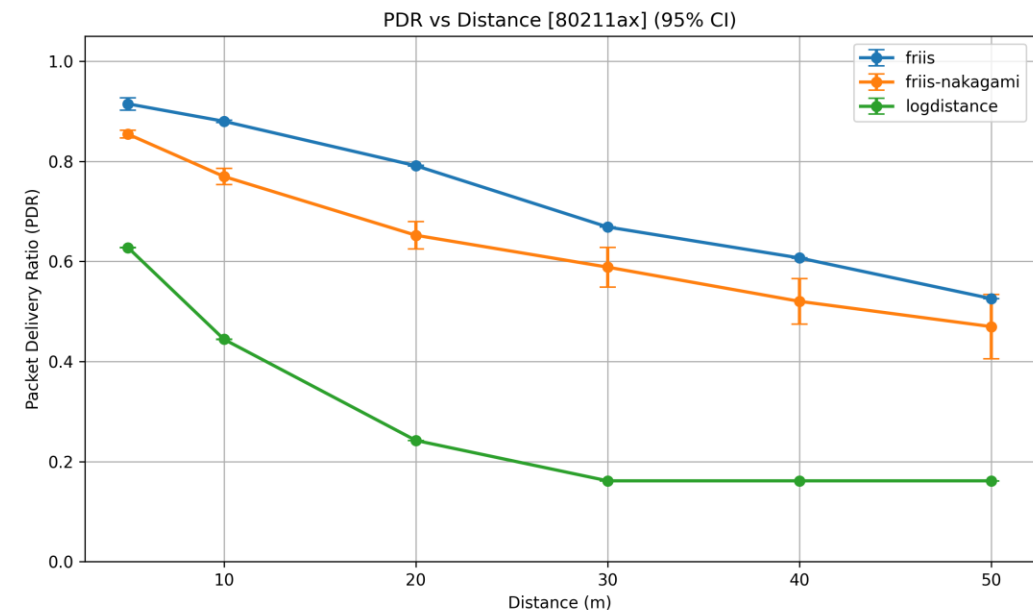
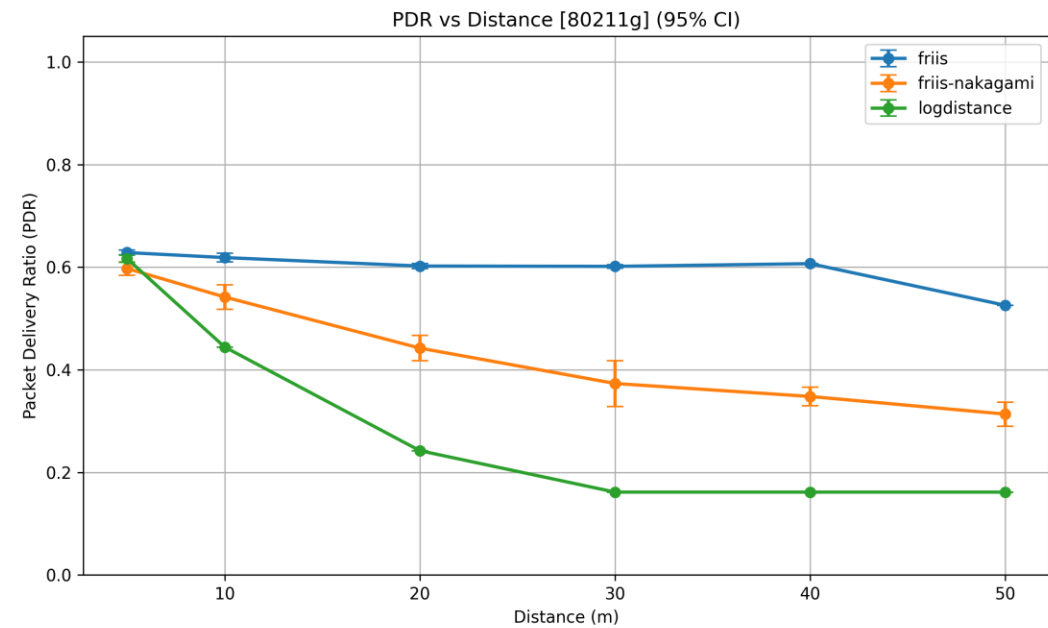
Conclusions

IEEE 802.11ax achieved superior overall performance in:

- throughput
- packet delivery ratio
- latency
- Wireless scalability degrades as:
 - node density increases
 - propagation conditions worsen
- Friis propagation overestimates practical performance.
- Friis-Nakagami provides more realistic fading behavior.
- LogDistance exposes severe attenuation and coverage limitations.
- PHY-layer improvements alone are insufficient to fully solve congestion and scalability limitations in dense WLANs.

Final Statement

- The obtained results demonstrate that advanced PHY-layer technologies such as IEEE 802.11ax significantly improve WLAN efficiency and reliability; however, propagation impairments, medium contention, and scalability constraints remain fundamental challenges in dense wireless environments.



THANK YOU

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Scalability, Reliability, and PHY-Layer Performance
Analysis in Dense Wireless Networks.



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